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Effects of Over-Time Exposure to Partisan Media and Coverage of Polarization on Perceived Polarization

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ABSTRACT

America is said to be more polarized than ever before, and extensive research examines the causes and effects of political polarization. A less studied but more pronounced trend is that citizens *perceive* society and politics as sharply divided. We focus on news exposure as a key reason for these perceptions. We disentangle the unique effects of exposure to partisan outlets and to news coverage of polarization in general, whether in partisan or centrist news media. We rely on 9 months of online behavioral data (127,400,000 visits) from large samples in the U.S. and Poland (total $N = 2,462$ & $2,120$), paired with a three-wave panel survey on participants' attitudes and perceptions. We measure individual exposure, over-time and unobtrusively, to partisan outlets as well as to news coverage of polarization through a fine-tuned BERT-based machine learning classifier. We find that both exposures increase perceived polarization, independently, in both countries. Yet, in both contexts, it is news coverage of polarization, regardless of whether it is ideologically left, right, or centrist news outlets, that is the strongest driver of perceived polarization. These changes are confined to personal-level polarization (perceived distance between oneself and members of the opposing party), while no measure significantly influences perceptions of society, more generally, as being polarized. These findings have important implications for our understanding of how the media's increasing tendency to focus on conflict as a central reporting frame can be the driver of often exaggerated perceptions of partisan divisions.

KEYWORDS

Perceived polarization;
media coverage; partisan
media; web browsing data;
computational social science

Over the past 50 years, the public in the United States has grown increasingly polarized along party lines. Some citizens, especially strong partisans, hold more extreme policy positions Gallup (2017) and express hostility toward out-party supporters (Iyengar et al., 2012; Mason, 2018). While by no means universal, these concerning trends have also carried

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over to other countries (Boxell et al., 2020). Although the extent of both policy and affective polarization is subject to some debate, there is largely undisputed agreement that many citizens *perceive* society as highly polarized (Westfall et al., 2015). In the U.S., for instance, individuals perceive more polarization than actually exists and perceived polarization has grown more rapidly than actual affective polarization (Enders & Armaly, 2019).

Given that these perceptions themselves have a range of negative effects, such as enhancing affective (Enders & Armaly, 2019) and attitude (Ahler, 2014; Ahler & Sood, 2018) polarization, decreasing trust (Lee, 2022) and efficacy (Enders & Armaly, 2019), and even generating depression and anxiety (Nayak et al., 2021), it is important to identify the causes of *why* the public sees the polity as sharply divided. In this project, we focus on exposure to (a) (partisan) news media and (b) news coverage of polarization as potential drivers of perceived polarization in two countries, examining their independent and interactive effects.

News media, which are the primary means by which citizens learn about society and elite behavior, emerge as a key culprit in heightening perceived polarization. Partisan news resort to the rhetoric of outrage (Berry & Sobieraj, 2014) and attack the opposition, and news media in general increasingly focus on polarization and political conflict (Levendusky & Malhotra, 2016), as we detail below. Exposure to such coverage exacerbates perceived polarization within the mass public (Levendusky & Malhotra, 2016; McLaughlin, 2016).

Nevertheless, it remains unclear whether it is partisan news or the coverage of polarization in news media, whether centrist or partisan, which makes the public perceive polarization. Further, in disentangling these two dynamics, are these effects additive or multiplicative? For instance, encountering coverage *about* polarization in partisan outlets may have an especially strong influence on perceived polarization. To our knowledge, no past evidence disentangles these two potential sources of perceived polarization using observational data on actual over-time exposure to online news and to the coverage of polarization wherein. Additionally, even less is known about these dynamics outside of the U.S., as the solid majority of work on affective and perceived polarization comes from this one country.

We rely on data from a large three-wave project that combines survey self-reports with online behavioral data from adults in the U.S. ($N = 2,462$) and Poland ($N = 2,120$). Every 3 months, the same participants submitted their online browsing data via our open-source web browser extension, Web Historian (see Supplementary Materials [SM] A.1) and also completed 15-minute surveys about their attitudes, perceptions, and behaviors. We rely on these behavioral traces of exposure (more than 127.4 million web visits) in conjunction with a comprehensive list of news domains (see SM D) and classifications of ideological leaning of news sources to identify partisan news. Furthermore, we train a neural binary classifier which identifies whether the specific news content that participants actually visited online across 9 months covers polarization and political divides (for details and model performance see SM E).

These longitudinal data – self-reports at three time points combined with quantified exposure to partisan media *and* to news coverage of polarization – allow us to predict over time variations in perceived polarization from exposure to (1) partisan and centrist domains as well as (2) online news about polarization and to disentangle the unique influences of each by studying each side-by-side.

Our findings are rather consistent across the U.S. and Poland. Although online exposure to news coverage of polarization is limited, accounting for an average of 14.8% and 8.2% of

participants' classified news visits in the U.S. and Poland, respectively, greater exposure to such news increases how distant participants see *themselves* from their political opponents (Perceived Personal Polarization), without affecting their perceptions of the polity as divided (Perceived Societal Polarization). These patterns are consistent across the countries, with exposure to coverage of polarization producing the greatest effects and exposure to partisan left and right media producing smaller but still significant effects. These overall effects are additive but not interactive, with exposure to news coverage about polarization being a consistent predictor regardless of the ideological slant of the reporting news media outlet.

(Partisan) News Media and Perceptions of Polarization

Many observers blame partisan media for political polarization in America, focusing on their effects on attitude or affective polarization (Guess et al., 2021; Levendusky, 2013; Wojcieszak et al., 2021). There are reasons to suspect that exposure to partisan news should also exacerbate *perceived* polarization. Partisan media, by definition, favor one side, and so their coverage is positive toward the preferred party (Baum & Groeling, 2008). At the same time, partisan media also cover the opposition in a negative light (Hall Jamieson & Cappella, 2008; Pew Research Center, 2017), focus on its scandals or wrongdoings (Puglisi & Snyder, 2011), and use outrage discourse, namely “overgeneralizations, sensationalism, misleading or patently inaccurate information, ad hominem attacks, and belittling ridicule of opponents” (Berry & Sobieraj, 2014, p. 7). Relatedly, partisan news coverage relies on exaggerated and negative exemplars to depict opponents, selectively choosing and covering their negative or uncivil behaviors (Levendusky, 2013). These exemplars then come to represent the entire opposition, leading individuals to see its actions and beliefs as extreme and divisive (Levendusky & Malhotra, 2016).

Thus, partisan media help fuel a rise in partisan misperceptions about the opposing party, which in turn result in a sense of greater distance from the opposition than exists in reality (Wilson et al., 2020). Both Republicans and Democrats systematically mischaracterize out-party positions (Moore-Berg et al., 2020), with misperceptions increasing as individual strength of partisanship – and thus also exposure to partisan news – increase. Indeed, to the extent that the rise in perceived polarization has been particularly pronounced among partisans (Westfall et al., 2015), it is possible that their consumption of partisan news is to blame for these (mis)perceptions (Wilson et al., 2020).

Although these features of partisan media are said to emerge across the left and the right, with some scholars noting that “the same rhetorical techniques are deployed on both sides of the political divide” (Berry & Sobieraj, 2014, p. 8), they are not uniform across the ideological spectrum. Some suggestive evidence shows that Republicans' perceptions of Democrats are significantly more stereotypical and biased than Democrats' (Ahler & Sood, 2018), and that right-leaning media generally have stronger polarizing effects (Garrett et al., 2019). It is not clear, however, whether there are ideological asymmetries in partisan news effects on perceived polarization.

Media Coverage of Polarization

We also argue, as a central contribution of this project, that news media in general, whether partisan or centrist, can increase perceived polarization through the very act of reporting on polarization, partisan conflict, and political divides, what we refer to as “news about polarization” or “news coverage of polarization.” Content about polarization is prevalent in news as conflict is a well established journalistic value (Harcup & O’neill, 2017) and is often used a central frame in political reporting (Project for Excellence in Journalism, 1998; Semetko & Valkenburg, 2000). The media cover negativity and clashes between partisan elites and ordinary voters to a much greater extent than cooperation or bipartisan agreement (Morris & Clawson, 2005). The U.S., for instance, is often painted in terms of “red versus blue,” creating an ingrained frame of division (Fiorina et al., 2005; Rutchick et al., 2009; Seyle & Newman, 2006). As shown by Levendusky and Malhotra (Levendusky & Malhotra, 2016), the word “polarization” has become more frequent in newspaper coverage in recent years.

Moreover, media content itself has become more polarized, especially since the rise of cable TV (Martin & Yurukoglu, 2017). Although, the rhetoric of outrage or incivility is especially prevalent in partisan TV, political blogs, or newspaper columns, the controversies are often picked up by centrist news simply because they can generate greater viewer attention (Berry & Sobieraj, 2014). In recent years, established media outlets have become more extreme in their content, featured fewer moderate contributors (Kim et al., 2022), and presented “in your face” debates, which lead the audience to see the opposition as not legitimate (Mutz, 2007).

These trends are heightened by the fact that parties themselves have become more polarized and political elites have increasingly turned to polarizing and uncivil language in their public communications (Ballard et al., 2022, 2023; Heseltine, 2024; Heseltine & Dorsey, 2022). Because journalists draw news stories from elite messaging (Van Aelst & Walgrave, 2016), they mimic these polarizing frames in their reporting, no matter how balanced this reporting may be. Relatedly, journalists use social media platforms as a source of news, relying on social media users to represent public opinion and illustrate what “an average citizen” is thinking (McGregor & Molyneux, 2020). As social media users are more polarized than the general population, journalists are amplifying the most extreme voices in society (McGregor, 2019). Exposure to such coverage in news may further exacerbate individual perceptions that the elites and partisans are extreme and that the polity is sharply divided.

Survey self-reports analyzed by Ahler and Sood (Ahler & Sood, 2018) suggest that following “what’s going on in government and public affairs” is correlated with perceiving out-party members in stereotypical ways, an important facet of perceived polarization. Additionally, relying on experimental designs, McLaughlin (McLaughlin, 2016) shows that a news story mentioning political conflict increases perceived conflict, and Levendusky and Malhotra (Levendusky & Malhotra, 2016) find that exposing subjects to news about polarization results in a small increase in perceived polarization. It is unclear whether these findings generalize to the “real world.” By studying exposure to a neatly isolated piece of content, which individuals may not encounter outside of the experimental setting,

existing studies may overestimate the effects of news coverage about polarization on citizens' perceptions.

Importantly, given that both partisan media and coverage of polarization in and of itself, regardless of the ideology of the source, may increase perceived polarization, we do not know which of these factors is the primary driver and whether these factors act in a supplementary, duplicative, or even interactive fashion.

This project disentangles the over-time influences of both – actual exposure to partisan news media on the left and the right and also to news coverage of polarization on people's perception of the polity as polarized. We examine each factor side by side – to our knowledge, the first undertaking of this kind, doing so in two polarized contexts: in the oft-studied U.S. and also in Poland, a newer and backsliding democracy in the European Union. We expect that higher levels of exposure to partisan news on the left and the right will predict increased citizens' perceptions that the opposition is more distant from them as individuals on policy positions (perceived personal polarization, H1a) and perceptions that the polity is sharply divided (perceived societal polarization; H1b). We have similar expectations about news coverage of polarization, whether in centrist or partisan media. Higher levels of exposure to news coverage of polarization should predict increases in perceived personal (H2a) and societal polarization (H2b). We also suspect that higher levels of exposure to news coverage of polarization in partisan media specifically will *interactively* predict increased perceptions of polarization over-time, such that the strongest increases in perceived personal and societal polarization will occur when individuals encounter news coverage of polarization as part of a more partisan rather than centrist media diet (H3a-b).

Lastly, we examine whether it is exposure to partisan news on the right or on the left that most strongly predicts perceived polarization (RQ1), and whether the potentially amplifying effects of exposure to news coverage of polarization within the right versus left partisan news have particularly pronounced effects (RQ2).

Context

This project focuses on the United States and Poland. As aforementioned, most extant work on perceived polarization, including the majority of work cited in this paper, focuses on the U.S., and research in other contexts is very limited. As such, expanding this focus can offer insights into whether the theorizing and evidence from the U.S. apply to other political and media systems. The U.S. and Poland provide an ideal test of how both partisan media and coverage of polarization shift perceived polarization as both countries have strong and clearly delineated partisan systems and a robust partisan media ecosystem. The U.S. is an often studied case, a two-party system with a liberal media system dominated by partisan cable channels (Hallin & Mancini, 2004) and with growing hostilities among the citizens. Partisanship is a salient political identity and forms the core of divides among the elites and the public.

Poland, in turn, is a post-communist and newer democracy, with liberal-polarized pluralist media system and public service broadcasting, which – until the parliamentary elections on October 15, 2023—was controlled by the far-right government. Although Poland is a multi-party system, with over 50 registered parties represented individually or in coalitions in elections, the core axis of divides emerges along one's support for or opposition to the government, which is linked with ideology and partisanship.

Specifically, the previous right-wing Law and Justice Party (PiS), which had been in power for 8 years until the last election, “has managed to reformulate the main dividing line of political life and turn its followers and critics into two unyielding hostile camps” (Fomina, 2019, p. 126), i.e., the supporters of the Law and Justice Party (PiS) and the opposition, including the Civic Platform Party, which won the 2023 elections jointly with the Civic Coalition political alliance. These two camps clearly differ in their underlying ideological and policy differences. Law and Justice Party (PiS) supporters have ethnic and religious (exclusivist) view of Poland’s identity, and support policies consistent with this view (e.g., anti-immigration, anti-minority rights, etc.). In turn, the Civic Platform Party and the Civic Coalition alliance at large have a non-exclusivist view of being both Polish and European, and disagree with the previous Law and Justice-led government on nearly all policies (Fomina, 2019). These conditions create (a) tangible perceptions of polarization within the public and (b)) relatively high potential for individual exposure to partisan media and to news media coverage reporting polarization.

Methods

We rely on a three-wave panel study fielded in 2019,¹ with respondents recruited via Lucid in the U.S. and Panel Ariadna in Poland. We enforced quotas on age, gender, and education (and ethnicity in the U.S.), with the samples intended to match national census margins. The first wave consisted of 2,462 respondents in the U.S. and 2,120 in Poland, with 1,261 and 976 completing the second wave, respectively, and 948 and 424 the third wave. In each wave and country, samples remained well balanced with little differential attrition over time. We offer full demographic information on the participants across the waves in each country in SM B.3.

In each wave, before taking the survey, participants were directed to Web Historian, an opensource tool that accesses respondent’s browser history stored on their computers (up to 90 days) and displays it to them using visualizations (e.g. network graph of websites visited, word cloud of used search terms, searchable table of browser history). After reviewing their browsing data, participants could eliminate the domains and search terms they preferred not to share (e.g., adult sites) and submit the data to the study following an extensive informed consent. Section A.1 shows screenshots of the interactive informed consent process.

We rely on 9 months of trace data from each country. We use trace data from the 90 days before Wave 1 to construct Wave 1 measures, data from the approximately 90 days before Wave 2 to construct the Wave 2 measures, and data from the approximately 90 days between Wave 2 and Wave 3 to construct Wave 3 exposure measures. In total, the final browsing data contained 88,266,597 visits in the U.S. and 39,173,024 in Poland, of which just over 1,400,000 were to news domains in the U.S. and 1,000,000 in Poland.²

The key *independent variables*, exposure to partisan news and to news coverage of polarization, were computed using these browsing data. We first calculate how often participants visited news over the three wave period by matching the visited domain (e.g., nytimes.com, foxnews.com) to identifiable news domains per country. In the U.S., these domains are compiled in a comprehensive list of 5,400 local and national news domains (of which 2,050 were visited at least once by our participants). In Poland, we compiled a total of

298 local and national Polish news organizations (291 visited). The details on how the lists were created are shown in SM D).

Having extracted news visits at the individual level, we identify visits to partisan domains for each participant relying on domain-level ideology scores. In the U.S. we use scores from (Robertson et al., 2018). These scores are constructed for media outlets based on the sharing trends of over 500,000 partisan Twitter users in the U.S., matched to voter registration data. These scores were extensively validated (see Robertson et al. (2018)) and highly correlate with classifications from other work (e.g., $r = .98$ with scores by Eady et al., 2019). These scores range from -1 to 1 , with -1 being the most liberal and 1 being the most conservative. We classify each news visit as either left, right, or centrist exposure based on scores of less than -0.2 being left, greater than 0.15 being right, and scores in between being classified as a centrist media.³

In Poland, in the absence of ideology scores for a comprehensive range of outlets, we construct our own scores in a fashion similar to U.S. scores. We first collected all tweets sent by Polish Members of Parliament and Senators between 2009 and 2022 (4,000,000). We then assign each Polish politician an ideology score based on the ideology of their party (as scored by expert ratings in the 2019 Chapel Hill Expert Survey). We then extract all links from the tweets, assigning, at the tweet-level, an ideology score to a link matching the ideology score of the sender. Ideology scores are then averaged at the domain level and then standardized across a -1 to 1 scale. Subsequently, similar to (Robertson et al., 2018), domains shared more by left-leaning parties will have a score closer to -1 , while domains disproportionately shared by right-leaning parties will have a score closer to 1 . Right-leaning outlets are defined as those with a score of 0.275 or higher, while left-leaning outlets are defined as -0.265 or lower. After creating these scores, the overall distribution of scores, as they relate to individual outlets, as well as the left/right/center thresholds, were reviewed and approved by two outside experts in Polish politics.

In SM D, we present the details on both approaches, refer the readers to public GitHub repositories, and offer graphical representations of the ideological placements of the most frequently visited domains in the U.S. and Poland. Ultimately, for both countries, we then measure individual exposure to left, right, and centrist media as a *percentage* of all individual news visits.

Importantly, in addition to the slant of the domain, we also determine whether participants encountered coverage of polarization in the news articles they visited. Three trained annotators per country (native speakers familiar with local politics) manually labeled 2,887 news articles from each country for whether they mentioned polarization, political divides, clashes between the major political sides (e.g., Democrats and Republicans in the U.S., the government and the opposition in PL, the left and the right), whether elites or ordinary voters, and also used extreme or polarizing language (codebook and details on the coding procedure can be found in SM E). We used these annotated data to train a transformer-based (BERT) neural binary classifier of news coverage of polarization (Devlin et al., 2019); precision = 0.84 ; recall = 0.84 ; accuracy = 0.84 ; see SM E for details). We scraped the text for all unique news URLs where text was scrapable and applied our classifier to identify which articles visited by each participant covered polarization (see SM D for full information on scraping and parsing process). From this, we calculate exposure to news coverage of polarization for each participant in each wave based on the total number of news visits to news about polarization as a (logged) count variable.⁴ As a robustness check, we also

measure this exposure using the percentage of news that is classified as content about polarization, finding that the results hold using this alternative measurement; see SM G

As *outcome measures*, we test two facets of perceived polarization, or how one perceives oneself vis-a-vis partisans from the opposing political group (Perceived Personal Polarization) and one's perceptions of polarization in society (Perceived Societal Polarization). To measure the former, we build on a measure by (Enders & Armaly, 2019) to test the perceived ideological distance between individuals and their out-groups. We asked respondents how similar or different the views of out-group members (i.e., out-partisans in the U.S. and supporters/opponents of the government in

Poland) were to their own views on six issues (e.g., the economy, climate change, immigration, etc.). We assess Perceived Societal Polarization by asking about participants' agreement or disagreement with four statements about divides within society (e.g., polarization in America/Poland is greater than ever before). We then create an index, on a wave-by-wave basis, using the respondent's average rating across the battery, with each question measured on a 7-point Likert scale. SM B.1 presents the question wording and descriptive statistics for these measures.

Analytical Strategy

To examine our hypotheses and questions, we rely on a multilevel model with random intercepts specified at the subject level. Multilevel random intercepts models are best suited to estimate the tested effects while accounting for the nested structure of the data (subjects nested in waves), thereby controlling for the individual's typical level on the outcome variable. In terms of temporal order, each observation in our data has a value for the predictor (i.e., exposure) and a value for the outcome, with the former measured before the outcome (i.e., the browsing data capture the 3 months before the survey). As such, any association between exposure and the tested outcome of the same wave cannot be due to reverse causality. That said, we cannot exclude the possibility that the outcome of a former wave (e.g., wave 1 perceived polarization) influences the predictor of a later wave (e.g., wave 2 exposure to news coverage of polarization). In SM F, we address this problem by assessing wave-on-wave change using cross-lagged panel models, which allows us to control for possible bidirectional influences. Although these models offer additional robustness checks, they come with their own limitations and may not be best suited for the limited number of waves we present (Hamaker et al., 2015). Furthermore, in SM F, we present justification for why we do not use fixed-effects models, often used for panel data. Specifically, the average within-unit standard deviation of our key measures across waves is very small, suggesting that random effects are the most appropriate modeling strategy based on the trends in our data and that we can have greatest confidence in between subject within-wave effects, as opposed to over-time causal effects, an issue we also address in the discussion.

In short, in the main text, we present estimates from the multilevel models with random intercepts, which is particularly useful when within-unit variation (i.e., individual change over time) is small. Specifically, we regress in two separate models, 1) perceived personal polarization and 2) perceived societal polarization, both measured for each individual on an indexed 7-point score, on the logged total of coverage about polarization that an individual viewed in a given wave, as well as the percentage of news visits to outlets classified as left-leaning and the percentage of news visits to outlets classified as right-leaning. We also

include individual-level demographic and political controls (i.e., age, gender, education, political interest, ideology, and partisanship) and also control for the overall number of (de-duplicated) news visits for each person/wave (i.e., a count variable of overall news visits) to isolate the effects of the specific exposures tested (i.e., partisan news and news coverage of polarization) from a person’s overall browsing.

To account for general shifts between waves across respondents (related to ongoing political developments, which may increase aggregate news consumption and aggregate perceptions of political polarization in a given wave), wave-level fixed effects are also included in the model. In using a mixed effects approach, the models control for both general fluctuations in perceptions and content exposure across waves for all respondents. These models include respondents who completed the final survey wave, thereby ensuring that all analyzed respondents are present in multiple waves.⁵

Trends in Exposure

We first report the descriptive patterns of exposure to partisan news and news coverage of polarization among our samples. First, the vast majority of the content that individuals view online is not news (just 2.28% of all the visits recorded in our samples were to news domains), and an even smaller percentage is visits to partisan media. As Figure 1 shows, in both countries, across waves, most news visits were to centrist outlets and visits to left-leaning outlets were more frequent than those to right-leaning outlets. Although these percentages depend on our binning of ideology, we note that if we look at the continuous distribution of visits, they also concentrate around the center. This concentration is also

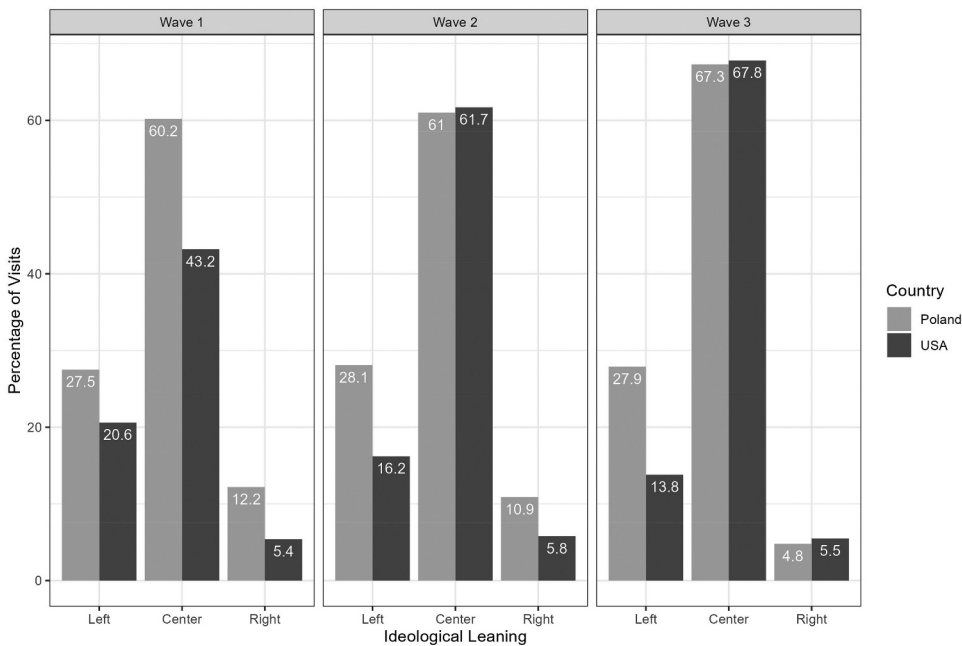


Figure 1. Percentage of news visits to partisan and centrist media. Grouped bar plot, divided by wave and country, showing that most news visits are to centrist outlets in both the U.S. and Poland.

consistent with past work (Guess et al., 2021). In Poland, over 60% of visits were to centrist outlets and roughly 27% were to left-leaning outlets in each wave. In the U.S., centrist visits increase as a proportion of all visits over time, rising from 43% in Wave 1 to 68% in Wave 3, potentially due to the growing controversies regarding Donald Trump and the impeachment hearings during the analyzed time frame, which may have prompted individuals to turn to centrist media for reliable information about these unfolding events. Figure 1 shows the percentages of partisan news visit relative to all news visits, regardless of whether a given news domain had an ideology score. In addition, SM B replicates these patterns with percentages taken relative to visits that received a classification from (Robertson et al., 2018).

More importantly, given our focus, Figure 2 shows fluctuations in exposure to news coverage of polarization. Despite the general increases in attention to polarization in American news media overall (Levendusky & Malhotra, 2016), actual exposure to news content about polarization was limited among our samples, accounting for 14.6% of all classified news visits in the U.S. and 8% in Poland. That is, news coverage of polarization constituted a minority of all news visits in each wave and across outlet ideologies even though – arguably – it represented a non-trivial percentage (ranging between 11.5% and 22.6% in the United States and 2.7% and 10.2% for Poland).

Over-time, exposure to news coverage about polarization did fluctuate between waves, especially within partisan news outlets in the U.S. Although news coverage of polarization among rightleaning media was at its highest in the U.S. in wave one, it increased in left-leaning media during waves 2 and 3, likely due to those outlets more

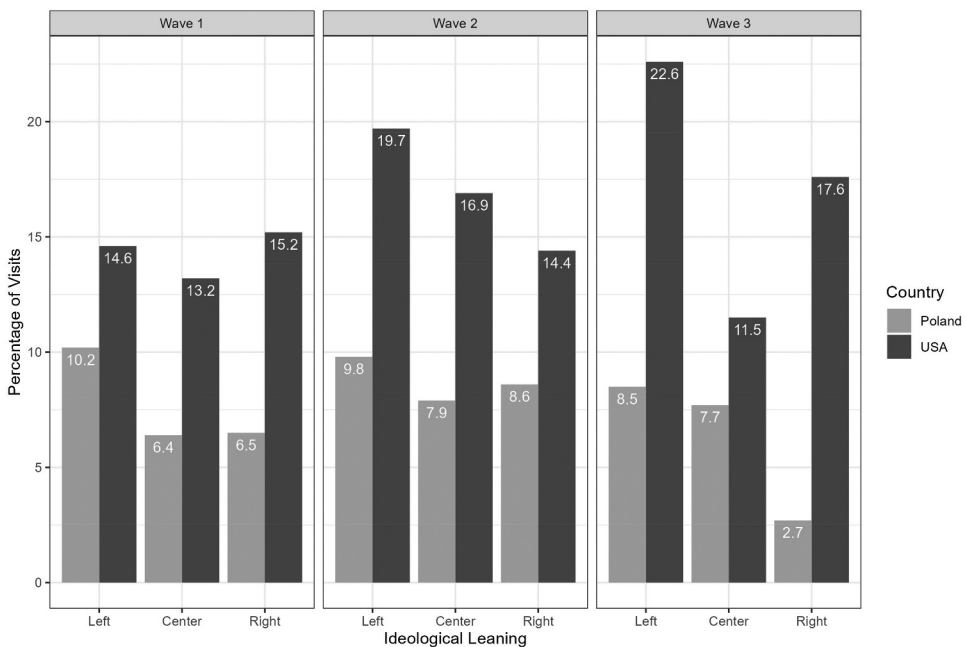


Figure 2. Percentage of news visits to coverage about polarization. Grouped bar plot, divided by wave and country, showing that the U.S. has more visits to news coverage of polarization and that more news content about polarization is found in left-leaning news media.

intensely covering Donald Trump’s impeachment proceedings during this period. In Poland, on the other hand, coverage of polarization in partisan media generally declined across the waves, with the third wave seeing the most substantial drop among right-leaning news outlets. This is likely because those pro-governmental outlets are try to create the illusion of public consensus and minimize public perceptions of the existence of the opposition and of public divides, and thus less likely to cover the opposition to the government than the left-leaning media. In SM D, we also provide a breakdown of coverage of polarization as a percentage of all visits at the domain level.

To more closely illustrate the dynamic fluctuations in news coverage of polarization across time, [Figures 3 and 4](#) plot the percentage of daily left-leaning, centrist, and right-leaning news visits classified as exposure to coverage of polarization, with horizontal lines marking two key political events in each country. In the U.S., the first line represents Trump’s much-covered comments in July telling Democratic Members of Congress to “go back” to where they came from, and the second represents the breaking of the whistleblower story which became the beginning of the impeachment hearings. In Poland, the first line corresponds with the Białystok Equality March in July 2019, which saw clashes between Pride marchers and far-right groups, while the second line shows the Polish parliamentary election in October 2019. For visual clarity, daily trends are shown on a 5-day rolling average.

The figures show, first, that the U.S. has greater general fluctuations in coverage of polarization than Poland. In the U.S., while the percentages of visits to coverage of polarization fluctuated across all outlets, it is the left-wing media visits which exhibit the most pronounced fluctuations. After Trump’s “go back” comments,

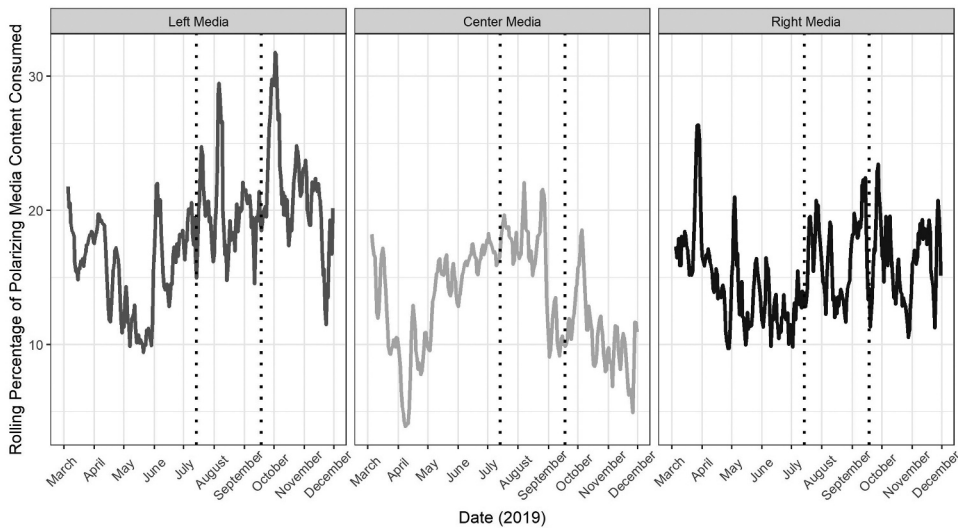


Figure 3. USA: rolling percentage of daily news visits to coverage about polarization, among classified visits, divided by outlet ideology. Line plot, divided by outlet ideology, showing that the percentage of news coverage of polarization seen in daily visits to left-wing media was most sensitive to breaking news developments in the U.S.

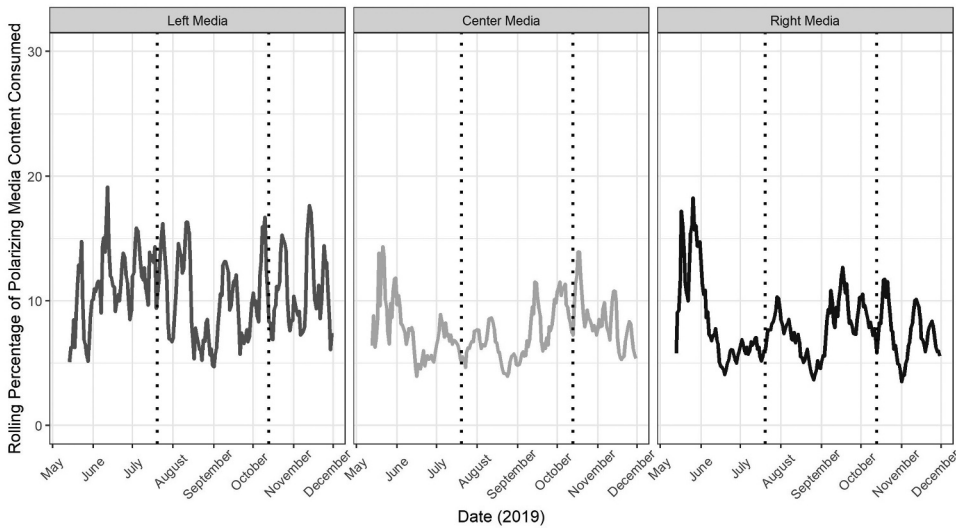


Figure 4. Poland: rolling percentage of daily news visits to coverage about polarization, among classified visits, divided by outlet ideology. Line plot, divided by outlet ideology, showing that the percentage of news coverage of polarization seen in daily visits to left-wing media was most sensitive to breaking news developments in the U.S.

which many on the left decried as racist, the percentage of visits to news coverage of polarization in left-leaning media shows clear increases. Similarly, after the Ukraine whistleblower story broke, there was an immediate increase in visits to coverage of polarization in left-leaning outlets, with elevated levels continuing throughout the impeachment proceedings. There were some fluctuations in visits to news coverage of polarization in the centrist and, to a lesser extent, right-leaning outlets.

In Poland, across all media types, coverage of polarization constituted roughly 10% of all coverage over time, generally fluctuating between 5% and 15%, regardless of outlet ideology. Relative to the key events, the level of change is minimal, although visits to coverage of polarization in right-leaning media visibly increased after the Equality March, and there was also an increase in visits to coverage of polarization across all outlet types in the buildup to and immediately after the parliamentary election.

Exposure and Perceived Polarization

We now turn to our hypotheses and research questions. Figure 5 shows the relationship between our two main outcome measures from exposure to left- and right-leaning media and of exposure to news coverage of polarization (as a logged count of visits). All measures are standardized so that effect sizes represent the effects of a one standard deviation increase in a given variable. Full regression models are shown in the SM. Models 1 and 2 show the results for partisan media and news coverage of polarization, while models 3 and 4 additionally include an interaction between left- and right-leaning media exposure and exposure to news coverage of polarization.

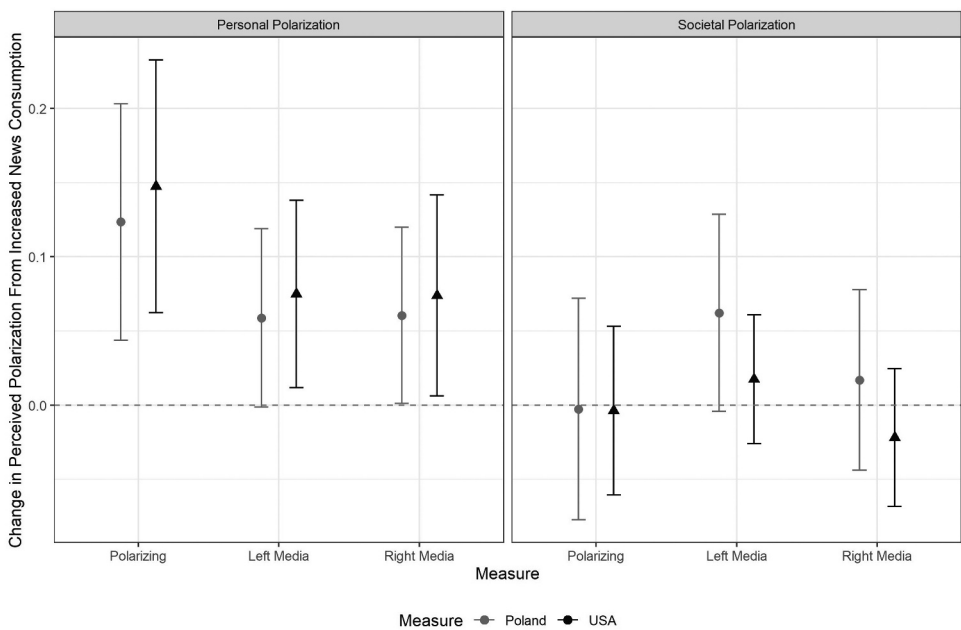


Figure 5. Mixed effects models predicting increases in perceived polarization based on changes in media exposure. Effects based on a (logged) count of visits to coverage of polarization and the percentage of news visits which are to news media classified as left- or right-leaning. Coefficient plots showing that partisan news exposure and also exposure to news coverage of polarization increase perceived personal polarization in the U.S. and Poland. All effects on perceived societal polarization are statistically insignificant.

Aligning with our theoretical expectations, greater exposure to partisan news media predicts increased perceived personal polarization (i.e., one’s perception of how distant others’ positions were from their own). This pattern emerges for both left- and right-leaning media in both countries, with the effect of exposure to left-leaning media in Poland having a p-value of 0.055. Addressing RQ1, no partisan asymmetries emerged; exposure to left- and right-leaning media exerted very similar effects.

Above and beyond, exposure to news coverage of polarization – regardless of whether in partisan or centrist outlets – also predicted greater perceived personal polarization in both countries. The standardized coefficients indicate that the effects of increases in exposure to news coverage of polarization are more than twice the magnitude of the effects of exposure to partisan media. In short, although both partisan outlets as well as news coverage of polarization each contribute to heightened perceptions that the opposition is more distant from oneself, in both countries it is mere exposure to news coverage *about* polarization which is the strongest predictor.

Contrary to expectations we do not find any effects on perceived societal polarization (i.e., one’s perception that the polity is divided).⁶ In standardized terms, coefficients for all measures, aside from left-wing media in Poland, were close to zero and did not approach statistical significance, suggesting a fairly consistent null effect on perceived societal

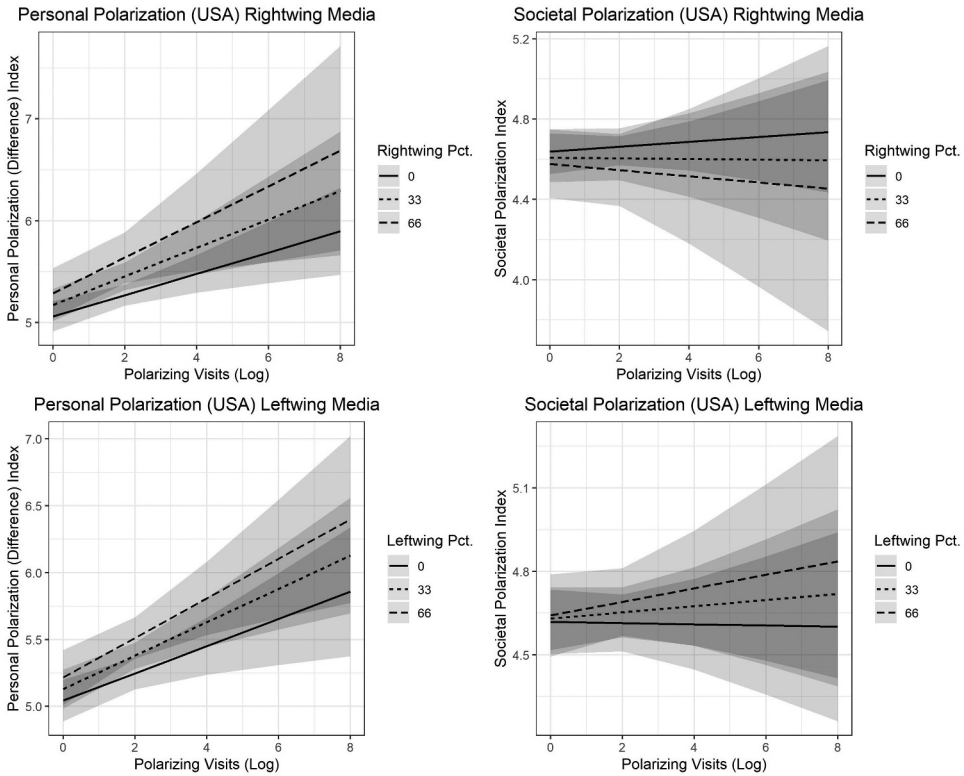


Figure 6. Interaction effects between partisan news media and exposure to news coverage of polarization (U.S.). Four interaction plots showing no statistically significant effects on perceived polarization from the interaction between the political leaning of news media and exposure to news coverage of polarization in the U.S. context.

polarization. Similarly, all robustness checks using alternate regression specifications and cross-lagged models reported in SM G and SM F.1 show identical null results for this outcome.

Finally, testing for interactive effects (Figures 6 and 7) suggests that while both exposure to partisan media and to news coverage of polarization have *additive* effects on perceived personal polarization, there is no evidence of an *interactive* relationship. Across both measures in both countries, the interaction between these two factors produces no statistically significant effects. That is, regardless of the slant of the news outlet, whether left, right, or centrist, encountering coverage of polarization has consistent effects across visits to all news domains.

Discussion

Perceived polarization in the United States is on the rise. Citizens' perceptions in and of themselves can exacerbate divides between opposing factions, erode political trust and efficacy, and also decrease individual well-being (Enders & Armaly, 2019; Lee, 2022; Nayak et al., 2021). In fact, individual perceptions may be more consequential than the

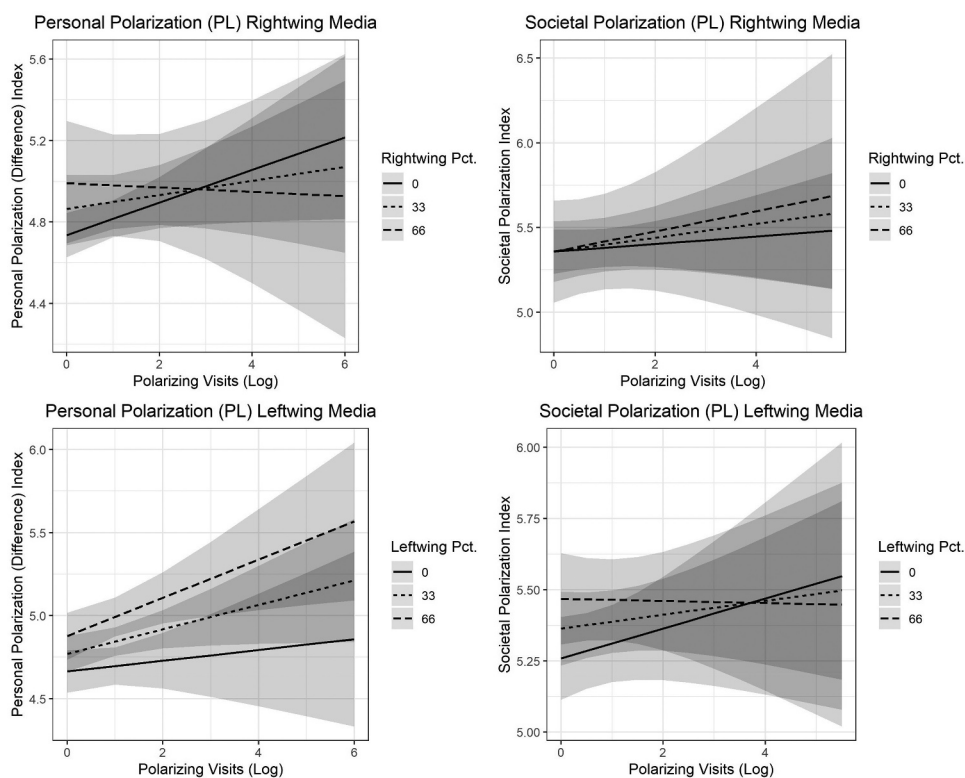


Figure 7. Interaction effects between partisan news media and exposure to news coverage of polarization (PL). Four interaction plots showing no statistically significant effects on perceived polarization from the interaction between the political leaning of news media and exposure to news coverage of polarization in the Polish context.

political reality (Wojcieszak & Price, 2012). In this project, we relied on over 127 million behavioral data points from over 4,500 individuals in two countries, used our extensive open-source list of news domains, and applied a validated neural binary classifier to identify news content of polarization in order to investigate the sources of the widespread perception that the polity is divided and that out-party supporters are distant from an individual on policy positions. We aimed to disentangle whether it is exposure to partisan news and/or mere exposure to coverage of partisan conflict that more strongly predicts these two facets of perceived polarization.

We offer four key findings. First, descriptively, we show that although news coverage of polarization constitutes a minority of all content seen by news consumers, exposure is far from insignificant and accounts for as much as 30% of daily news exposures in the U.S. and 18% in Poland. In addition, the prevalence of this coverage does fluctuate based on breaking news developments at any given time. As such, news consumers are regularly exposed to this type of content in news media, especially in the U.S., especially in partisan media, and especially at times when external events are indeed suggestive of strong divides within the political system.

Second, we find that exposure to news content about polarization in partisan as well as centrist outlets predicts increased perceived personal polarization. In other words, the more

citizens encounter news coverage of inter-party clashes, incivility between political factions, and language that clearly denotes polarization, the more they will see their outgroup's policy positions as distinct from and oppositional to their own. This perception may ultimately lead people to withdraw from interparty conversations and decrease whatever possibilities for contact across group divides.

Third, we also show that greater exposure to partisan media on the left and the right in both the U.S. and Poland predicts higher perceptions of how distant out-partisans are from an individual. This pattern emerges over time and in naturalistic settings of actual online exposures to a wide range of partisan websites. Furthermore, this pattern is *not* attributable to the fact that partisan news media are more likely to report on inter-party clashes than centrist news media. Because we tested exposure to both partisan news and to news coverage of polarization side-by-side in one model, we can conclude that it is the partisan cues, the nature of the reporting, the frames and narratives, among other features of partisan media—*not* the coverage of polarization wherein – that are the primary predictor of these effects on viewers' perceptions. This finding extends past work on partisan media, showing yet another outcome that is negatively affected by exposure to partisan outlets. We also note that these increases in perceived personal polarization exist in an additive rather than duplicative or interactive fashion, such that it is *not* the case that increased exposure to news coverage of polarization in partisan media is especially strongly associated with perceived polarization.

Lastly, in contrast to our theoretical expectations, there is little evidence to suggest that citizens' perceptions of polarization at the societal level – i.e., seeing the polity as divided and the inter-party differences as too deep to reconcile – are predicted by either partisan news consumption or exposure to news coverage of polarization. This may in part be due to high levels of perceived polarization in the contemporary political environment creating a combination of inflexible perceptions coupled with general ceiling effects. Across all waves and in both countries, perceived societal polarization remained consistent and high (see SM B), while average ratings of personal polarization did vary over time, suggesting relative stability in public perceptions of societal polarization, regardless of temporarily confined events. Taken together then, these results suggest that personal perceptions of polarization are most affected by partisan media and news coverage of polarization, with overarching perceptions of general polarization at the national level proving less malleable.

Naturally, with only two countries studied, we cannot shed more systematic light on how various system and context-level factors (e.g., the number of parties, media freedom, GDP, among other country-level characteristics) influence the detected patterns. Nevertheless, our project expands the still limited evidence on polarization from the U.S. to an understudied country, showing that the central results are closely aligned despite the many differences between the U.S. and Poland. These similarities in the results may be due to the fact that Poland, although a multi-party system, increasingly resembles the two-party U.S., with similar (if not higher) polarization levels. We invite scholars to build on our evidence and rely on the developed classifications to replicate our findings in different – ideally less polarized – contexts. Accumulating knowledge from non-WEIRD (Western, Educated, Industrialized, Rich and Democratic) countries will contribute to comparative work and to the further internationalization of (political) communication research. For now, we believe that the benefits of expanding the focus of this scholarship – and of political

communication research in general – to other socio-political contexts is one of the core contributions of our results.

We acknowledge that, as with any study of this nature, our trace data cannot capture the entirety of individual news diets, such as news exposure offline or on mobile devices. As such, our results may offer relatively conservative estimates of the prevalence of exposure to partisan news and to news coverage of polarization, given that more people consume news in general and partisan news in particular on television than online (Muise et al., 2022). Thus, there are reasons to believe that our descriptive and predictive results may be amplified if non-browser-based media were included in the data. Extensions of this project that incorporate other platforms and modalities are an important endeavor for future research. Such extensions are also needed to reconcile some conflicting evidence. Extant research that finds associations between (partisan) news and polarization tends to focus on television and relies on (largely unreliable) self-reports (Prior, 2009). In turn, the research that uses online traces largely finds null effects of online (partisan) news exposure on polarization (Guess et al., 2021; Wojcieszak et al., 2021). Designs that aim to capture exposure offline (e.g., TV or talk radio) and online are needed to offer a more systematic portrayal of (partisan) news exposure. Yet, such designs are ever more challenging given that it is nearly impossible to account for all the different media, platforms, and modalities in the current hybrid media system.

We also note that our data (and any other web browsing data) can only capture individual more active engagement with online content, i.e., the instances when a user clicked on the URL and landed on a story, not the more passive exposure to content that occurs when users scroll through news portals or their social media feed. These realities of online data collection introduce noise. On the one hand, we are underestimating exposures to content about polarization (which users may see on portals or social media). On the other hand, we may be overestimating such exposures if some users click on a link but only skim through a headline without reading content.

To the extent that such cursory exposure may also influence perceived polarization, the noise in our (and similar) traces may lead us to underestimate the prevalence of exposure to news content about polarization and its effects. Sadly, existing methods of online behavioral data collection at scale cannot shed light on which situation – under- or over-estimation of certain exposures – is more likely.

Finally, we reiterate that the limited over-time variance in our data limits the over time causal conclusions that can be drawn from these results, with the data being most suited to the analysis of within-wave predictive effects. In general, in the absence of external shocks, individuals are unlikely to significantly alter their news consumption in a confined time period in a fashion sufficient for deeper causal over-time analysis. This highlights the general stability of individual news diets, in the aggregate while ultimately presenting methodological constraints on researchers, even in the presence of granular individual trace data.

The detected patterns also underscore the fact that numerous other factors influence citizens' perceptions of the political system and the divisions therein. These can be shaped by one's immediate environment, such as interpersonal conversations with friends, family, or at work, one's broader community context, e.g., whether politically homogeneous or diverse, among other factors that may not be directly related to exposure to partisan media or the coverage of polarization more broadly.

Given that individual perceptions do have effects and given that – in the case of perceptions of polarization – these effects can be negative for citizens and the democratic system at large, it is important to understand the role of news media in generating these perceptions. Most observers worry about the impact of partisan media on (affective) polarization. Our work extends these worries, showing that over-time exposure to partisan news online additionally predicts increases in perceived policy distance from one's opponents. Yet and more importantly, our work also suggests that it is the general news media focus on clashes between opposing views and inter-party hostility that is at least partly to blame for the growing perceptions of polarization. News media – centrist or partisan alike – have been criticized for their excessive focus on negativity and conflict (Levendusky & Malhotra, 2016). Our study shows that these critiques are valid. Such focus not only decreases individual well-being (Boukes & Vliegenthart, 2017), leads to frustration among the electorate, or makes citizens disillusioned with the political process (Jones, 2001), but can also lead citizens to exaggerate the extent to which their opponents are different. Yes, conflict and negativity sell and generate attention (Soroka et al., 2019; Yu et al., 2023). Yet, given the central role news media play in a democratic society, and consequently their responsibility, the commercially driven race for audience attention may be doing a disservice to citizens and the polity at large.

Notes

1. Wave 1 was run between April and July 2019, Wave 2 was run in August/September, Wave 3 in November/December.
2. In order to avoid over-reporting online news exposure based on activities such as refreshing a browser or reloading a news domain home page, we de-duplicate the data at the respondent-day-domain level, and so 1,400,000 and 1,000,000 reflect the number of de-duplicated news visits across our samples.
3. The average scored news visit sits just to left of 0, with the classifications therefore being left-adjusted with cutoffs sitting on the outside of the ideological rating of the nearest major news outlet. The data do offer face validity and align with past work (Eady et al., 2019; Wojcieszak et al., 2023). For example, known left sites like Huffington Post or Mother Jones are categorized as liberal, right-leaning sites like Breitbart or Fox News as conservative, and most legacy media sites as centrist (e.g. PBS, CBS News, Reuters).
4. As the central expectation is that more exposure to news about polarization will predict increased perceptions of polarization, but with the realistic expectation that this effect may not exist in a linear additive fashion, we do not use a simple count variable.
5. For perceived personal polarization, we analyze waves 1 and 2 because only these waves contained this measure. In SM G, we also show models that include all respondents who completed at least two waves.
6. In Poland, there is some suggestive evidence (P-value of 0.066) that left-leaning media may predict increases in perceived societal polarization. However, this coefficient does not cross the 0.05 in any of the robustness models.

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